

Developing a multimodal transport simulation model using a case study of the Suburban Rail Loop East, Melbourne, Australia

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ABSTRACT

Agent- and activity-based transport models when combined link transport network flows to the individuals generating them, enabling exploration of the interplay between travel needs and transport network capacity.

This study expanded a recently developed multi-modal agent- and activity-based transport model of Greater Melbourne, Australia by incorporating travel mode choices of demographic cohorts, a more accurate representation of public transport trips, and walking or driving to train stations. Separate mode choice models were developed for five unique cohorts of travellers. The model was calibrated and validated for each cohort using Victorian Integrated Survey of Travel and Activity data. The model simulated a Baseline scenario of existing transport infrastructure and an alternative scenario that includes Stage 1 of the Suburban Rail Loop, or SRL East, a new rail network traversing under-served middle suburbs. Results show SRL East will increase walking trips and average daily walking distances, and decrease car trips in areas surrounding SRL East train stations. The average walking distance between SRL areas increased between 4.9% and 11.1% for the respective cohorts, with the largest increase for those aged between 20 and 24.

Results suggest SRL East will change people's travel behaviour and influence the uptake of active transport, with the cohort results useful for age and gender-based interventions. The research is significant for showing SRL East's impacts on active transport up-take and for influencing city-wide transport and urban planning. Future research modelling local and city-wide impacts of all stages of the SRL are warranted.

1. Introduction

Walking and public transport are sustainable forms of mobility and integral to the efficient functioning of populous major cities, contributing to the mitigation of city-wide problems such as road network congestion, and air and noise pollution (Giles-Corti et al., 2016). They also contribute important co-benefits such as improved travel times and health since walking is an easily accessible form of physical activity with known physical and mental health benefits (Boone-Heinonen et al., 2009; Cerin et al., 2017; Hall et al., 2020). In large cities, many public transport users access train stations by private motor vehicle (Jeffrey et al., 2019; Vijayakumar et al., 2011). For example, in Greater Melbourne, Australia, as many as 80% of train commuting trips commence by car, particularly in middle and outer suburbs where distances

between residences and train stations are long, and areas are under-served by connecting bus routes (Jeffrey et al., 2019).

Transit-oriented developments (TODs) co-locate high-density residences and key destinations such as shops and services near transport hubs. TODs facilitate local living and access to regional opportunities including education and employment, reducing the need for cars (Bartholomew & Ewing, 2011). Consequently, TODs can alleviate concerns arising from population growth and the pressure this puts on road networks and destination accessibility. However, increasing accessibility by building new train lines and improving train station precincts can be expensive to deliver and their impact on travel behaviour is difficult to estimate before construction.

To assist, transport simulation models can be used as virtual laboratories to model planned infrastructure changes on travel behaviour

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and walking to public transport. These models can test the network-wide impact of changes in public transport infrastructure and services before real-world implementation (Schlenker et al., 2022). Many simulation modelling frameworks exist, with the most widely used being the traditional four-step transport modelling framework that captures mobility with aggregated trip flows (McNally, 2007). In contrast, activity-based modelling follows a bottom-up and disaggregated approach in recreating a transport system by connecting trips to individuals (McNally & Rindt, 2007; Rasouli & Timmermans, 2014). Key advantages of activity-based models are that they allow for investigation into individual level travel behaviour and can accommodate transport interdependencies such as multimodal trips with public transport, driving, cycling, and walking. Additionally, joining activity-based transport demand models with agent-based transport simulation modelling makes it possible to examine the impact of heterogeneity in agents' decision-making and inter-agent interactions as well as interactions with the environment (Kagho et al., 2020).

Agent- and activity-based models are being used globally for modelling transport behaviour and for measuring mobility-related impacts of city-wide road and rail network infrastructure developments (Bösch et al., 2016; Erath et al., 2012; Hörl & Balac, 2020; Infrastructure Victoria, 2017; Rieser, 2016; Ziemke et al., 2019). One of the most well-known and widely used agent- and activity-based simulation software platforms is the Multi Agent Transport Simulation (MATSim) toolkit (Horni et al., 2016), which has been used to study different aspects of transport systems and travel behaviour, from public transport and congestion modelling to novel modes of transport such as shared autonomous vehicles (Kamel et al., 2019).

A key advantage of agent- and activity-based models, such as MATSim, is their capacity to model detailed trip components, including walking and driving to or from a train station, at the individual level. This enables the investigation of local and city-wide travel behaviour changes using a simulation model. For instance, Manser et al. (2020) used MATSim to design an optimised public transport network for the city of Zurich, Switzerland, aiming for an economically profitable public transport system under two scenarios of subsidised and non-subsidised public transport. They measured various passenger and system level variables under the two scenarios. Leng and Corman (2020) used MATSim to model the change in individuals' routes when a major disruption in the public transport service occurs. As another example, Balac and Hörl (2021) used MATSim to model the potential of using docked bike sharing services as the last/first mile solution for public transport in the San Francisco Bay area in the US. Although these studies demonstrate the potential of a tool such as MATSim to model and analyse travel behaviour at individual, local and city-wide level, a key gap remains due to complexity and the need to calibrate these simulation models across the constituent modelling components and input variables.

MATSim employs a coevolutionary optimisation approach to assign modes to each traveller based on a utility function, as detailed by Horni et al. (2016). This utility function and its coefficients are typically derived from a statistical mode choice model. Consequently, while MATSim, as an agent-based model, can theoretically capture individual-level behaviours, the extent of heterogeneity it can represent largely depends on the variables and factors incorporated into the mode choice model. Thus, it is crucial to carefully consider the type and level of heterogeneity expected from a MATSim model when constructing and estimating the choice model intended for use within MATSim.

Several studies have enhanced mode choice models by incorporating socio-demographic or ecological factors. Among different socio-demographic attributes, age and gender are commonly found to be significant factors associated with mode choice (Boulange et al., 2017; Kroesen & Chorus, 2018). However, few studies have integrated these refined mode choice models into large-scale transport simulations. One notable example is the work of Müller and colleagues (Müller et al., 2021). They grouped the population into ten distinct subpopulations

based on socio-demographic characteristics such as age, gender, and income, and developed a distinct mode choice model for each subpopulation. However, they did not calibrate the transport behaviours for each individual subpopulation.

Jafari et al. (2024) developed an Agent and Activity Transport model of Melbourne (AToM), using MATSim as its core transport simulator. They used the activity-based travel demand generation model developed by Both et al. (2021) and a road network generation algorithm developed by Jafari et al. (2022) and a simple mode choice model for four travel modes of walking, cycling, public transport and car as the main inputs of their MATSim model. In their model, Jafari et al. (2024) employed only a simple mode choice model without demographic specifications, and walking was the only mode of transport considered for travel to or from a public transport station. Therefore, their model was not suitable to investigate changes in access and egress modes for public transport trips across different demographic cohorts of the population, which is needed for assessing the health and environmental impacts of constructing new major rail infrastructure.

In this study, we extend the Agent and Activity Transport model of Melbourne (AToM) of Jafari et al. (2024) to make it suitable for investigating both walking and car trips to or from train stations and for different demographic cohorts including age and gender across a large-scale city-wide simulation. This was achieved through two major extensions: (1) developing and estimating a cohort-based mode choice model that we used in the simulation and calibrated using real-world data, and (2) adding a train station and access/egress mode assignment step to the demand generation process, which was calibrated using train station usage data. Finally, we demonstrated the usability of the calibrated model by employing it to estimate mobility patterns and mode share changes resulting from a major rail network infrastructure project in Melbourne, Australia, known as the Suburban Rail Loop (SRL) project, which is referred to as a 'city-shaping' project (State Government of Victoria, 2021). The first stage of this project, also known as SRL East, is investigated in this study.

2. Overview of Agent and Activity Transport model of Melbourne (AToM)

AToM is a large-scale agent- and activity-based transport model for simulating the daily travel of synthetic individuals representative of the population of a city such as Greater Melbourne (Jafari et al., 2024). AToM uses the Multi-Agent Transport Simulation toolkit (MATSim) (Horni et al., 2016) as the transport simulator and uses two main inputs: activity-based transport demand for Greater Melbourne developed by Both et al. (2021) and a detailed transport network for Greater Melbourne developed by Jafari et al. (2022).

The activity-based transport demand used Victorian Integrated Survey of Travel and Activity (VISTA) survey data from 2012 to 2018, to construct daily travel plans for a synthetic population (State Government of Victoria, 2019). The synthetic population was derived using Australian Bureau of Statistics (ABS) Census 2016 data and contains individuals (ABS, 2016), each described by a set of age and gender demographic attributes, along with household membership and location. Daily travel plans comprise a sequence of activities generated for each synthetic person, where each activity is defined by its type, start and end time and duration, and are chosen based on their demographics. The plan also prescribes the location for each activity along with the transport mode used for travelling between them, with locations chosen based on the likelihood of a trip occurring for that distance for using the chosen mode and the activity-based attraction of destination sites (Both et al., 2021).

The virtual road network used OpenStreetMap (OSM) data for Greater Melbourne for March 2021. It includes a set of links representing road segments, and nodes representing intersections and roundabouts. This network contains both major and minor roads for driving, walking and cycling, and attributes that impact travel such as road type, speed

limit and capacity. Additionally, General Transit Feed Specification (GTFS) data, which provides public transport schedules and associated geographical information, was used to generate the public transport links, station/stop locations, and service timetables. The GTFS data used a schedule from mid-October 2019 (Jafari et al., 2022).

The AToM development workflow did not consider two transport behaviours: the relationship between demographic characteristics of travellers and their mode choice behaviour; and multimodal public transport trips. In this study and as shown in Fig. 1, we extended Jafari et al.'s (2024) AToM model to provide separate mode choice models for demographic cohorts based on age and gender and extended the model to include multimodal public transport trips to model walking and car trips from home to train stations and *vice versa*.

3. A cohort-based transport model for greater Melbourne

This section details the simulation model extension of AToM accounting for demographic heterogeneity in transport behaviour. Section 3.1 explains the development of the separate mode choice models for distinct age and gender cohorts. Section 3.2 explains the simulation model parameters and configuration of MATSim enabling the incorporation of access and egress trips to train stations explicitly into the workflow, better capturing the complexity of multimodal trips in Greater Melbourne. Section 3.3 describes how the model was calibrated to fit the data and validated.

3.1. Cohort-based multinomial mode choice models

Demographic attributes such as age and gender are important when modelling human behaviour related to transport mode choice (Bautista-Hernández, 2020; Grue et al., 2020). This study focused on age and gender as primary variables considering that these demographic attributes significantly influence travel behaviour patterns. Although a broader range of individual-level factors, such as socioeconomic position (e.g., occupation, education), and contextual built environmental factors (e.g., access to transport, shops and services, jobs) (Boulange et al., 2017; Charreire et al., 2021; Saelens & Handy, 2008) may also affect mode choice, they were not considered in this simulation model.

We used VISTA travel survey data from 2012 to 2018 in the simulation workflow with VISTA data from 2016 to 2018 used for model calibration (State Government of Victoria, 2019). Detail on the use of VISTA is provided throughout the following sections. Briefly, the VISTA travel survey is a multi-stage, variable-proportion, cluster sampling of households, with individuals who recorded one day of travel and activity across the State of Victoria, Australia including the city of Greater Melbourne (Both et al., 2021; Boulange et al., 2017). Table 1 illustrates the mode choice breakdown of the 64,500 VISTA 2012–2016 travel survey participants by age and gender groups. It shows for all groupings that driving by car is the most popular transport option, although public transport (PT) use is popular amongst participants aged 15 to 34.

The mode choice models were estimated on separate cohorts of travellers to classify travel activity. This approach allows for greater specificity in the simulation of travel activities for each cohort. The cohorts were determined using demographic-based clustering of travel behaviour patterns using Ward's method (Hair et al., 2014) following the approach of Both et al. (2021). Ward's method is a hierarchical clustering algorithm that groups data based on the similarity of input variables. It uses the sum of squares across all input variables to determine the clusters. Specifically, travel survey data from VISTA 2012–2018 were clustered based on the probability of engaging in work, shopping, personal, and social/recreational activities according to five-year age groupings by gender commencing at 0–14 years up to 65 plus years.

The resulting demographic characteristics of the five cohorts are shown in Table 2.

For each of these cohorts, a mode choice model was estimated using

multinomial logistic regression to provide estimated coefficients to capture travel behaviours according to age and gender and transport mode. This is an extension of the mode choice model used in AToM (Jafari et al., 2024), which used a single mode choice estimation for all demographic cohorts (Jafari et al., 2024). The choice of multinomial logistic regression for each cohort was influenced by its capability to yield detailed, distinct data for each cohort and their mode choices leading to more accurate modelling of transport mode choices. Moreover, these models are particularly compatible with the simulation framework employed in this study, offering a straightforward way of implementing the mode choice models compared to other model types, such as Latent Class models (Eshima, 2022). This will be further detailed in Section 3.2, where the use of MATSim as the simulation platform is discussed. In this platform, each cohort is seamlessly integrated and configured as a subpopulation within the travel demand model. The scope of variables (i.e. age and gender) included in the current study was influenced by data availability and the need to ensure sufficient and reliable data for each cohort in the analysis.

The mode choice model shown in Equation 1 was estimated separately for each cohort using multinomial logistic regressions to calculate the disutility of travelling by each of the four main transport modes of walking, cycling, public transport, and driving. In Equation 1, $S_{trav,i}$ represents the disutility of travelling by mode i , asc_i represents the alternative specific constant of mode i , $\beta_{trav,i}$ represents the marginal utility of travel time of mode i , $t_{trav,i}$ represents the travel time of mode i , β_m represents the marginal utility of money, and m_i represents the monetary cost for mode i .

$$S_{trav,i} = asc_i + \beta_{trav,i} \times t_{trav,i} + \beta_m \times m_i \quad (1)$$

where i = Drive, PT, Walk, Cycle

The alternative specific constant for driving was set to be zero in the model. To estimate travel time $t_{trav,i}$ for all modes, we used the Google Maps platform,¹ estimating travel time and distance for the travel mode used for each VISTA data trip from 2012 to 2016, along with all alternative travel modes that were not selected for that trip. For this, travel times were extracted from a mid-week workday in January 2022.

We assumed no monetary cost for walking and cycling trips. For public transport, a constant trip-based fare was used to represent the monetary cost argument, \$4.30 for each trip based on 2018 public transport fares for Greater Melbourne. For driving, m was calculated based on Equation 2, in which $d_{trav,drive}$ represents the distance travelled by car, 8.8017 (lit/100 km) is the average fuel consumption for an average vehicle in litres,² and \$1.16 is the average annual fuel price per litre in Victoria in 2016,³ chosen to match the timeframe of the VISTA data.

$$m_{driving} = \frac{\$1.16 \times 8.8017 \text{ (lit/100km)} \times d_{trav,drive}}{100} \quad (2)$$

After calculating the mode choice variables for the selected mode and all alternative transport modes, we estimated five multinomial mode choice models, one for each demographic cohort. Table 3 shows the results of the multinomial mode choice parameters estimated for each cohort. The alternative specific constants represent the influence of all unobserved factors and variables not explicitly included in the model, aside from travel time and travel cost. These constants capture mode-specific preferences that may affect mode choice. The β_{trav} coefficients represent the influence of travel time on the selection of different travel modes. The β_{trav} values are consistently negative in line with our expectation for a negative effect of travel time on mode choice. A larger

¹ <https://developers.google.com/maps/documentation/distance-matrix/overview>.

² https://www.atap.gov.au/sites/default/files/pv2_road_parameter_values.pdf.

³ <https://aip.com.au/aip-annual-retail-price-data>.

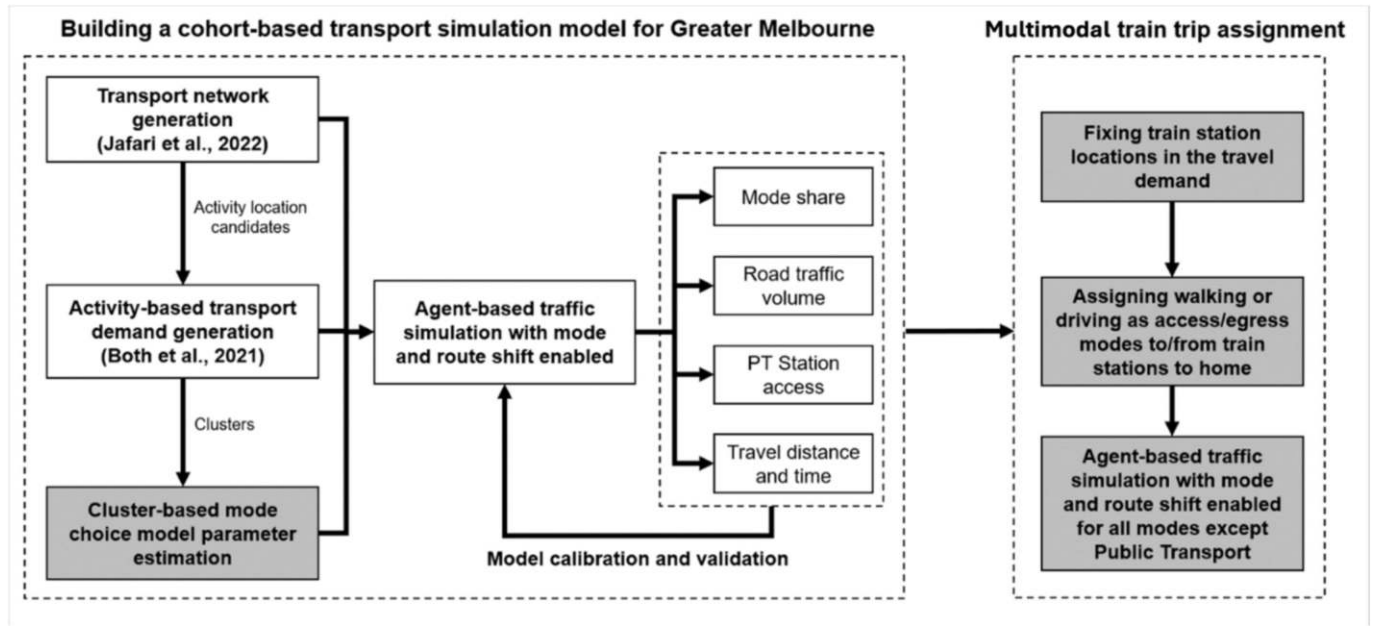


Fig. 1. Workflow diagram showing AToM, with modelling extensions highlighted in grey.

Table 1

Mode choice likelihood of VISTA travel survey participants along with population count.

Age group (years)	Female mode choice (%)				Male mode choice (%)				Population (n)		
	Walk	Bike	PT	Car	Walk	Bike	PT	Car	Female	Male	Total
0–14	17.0	2.1	6.0	74.9	18.3	2.1	6.7	72.9	4484	7103	11,587
15–19	15.9	0.6	23.8	59.7	17.6	3.2	22.0	57.2	1538	2262	3800
20–24	15.7	1.4	16.8	66.1	15.7	2.2	14.7	67.4	1528	2312	3840
25–29	20.4	2.4	13.9	63.3	19.3	3.7	12.4	64.6	1684	2330	4014
30–34	20.2	1.3	10.4	68.1	20.0	2.7	11.6	65.7	1866	2538	4404
35–39	19.0	1.2	7.5	72.3	16.7	2.6	8.3	72.5	1862	2722	4584
40–44	15.5	1.0	5.8	77.7	16.8	1.8	7.5	73.9	1992	2742	4734
45–49	14.7	0.6	4.9	79.8	15.1	2.4	6.5	76.0	1891	2607	4498
50–54	16.4	0.6	6.2	76.8	16.9	1.9	5.6	75.6	1895	2814	4709
55–59	16.9	0.6	5.1	77.4	14.7	1.4	5.6	78.2	1813	2629	4442
60–64	17.5	0.9	6.1	75.5	16.1	1.4	4.3	78.3	1648	2280	3928
65 plus	17.8	0.7	5.2	76.2	18.4	1.3	4.2	76.2	4194	5766	9960
Overall	17.3	1.2	8.3	73.2	17.4	2.1	8.2	72.3	26,395	38,105	64,500

Table 2

Demographic cohorts.

Name	Demographic characteristics
Cohort 1	Female and Male 0–19
Cohort 2	Female and Male 20–24
Cohort 3	Female 25–29 Male 25–59
Cohort 4	Female 30–59 Male 60–64
Cohort 5	Female ≥ 60 Male ≥ 65

Table 3

Estimated parameters of the multinomial logistic regression mode choice models.

Parameter	Estimated coefficients for each cohort				
	Cohort 1	Cohort 2	Cohort 3	Cohort 4	Cohort 5
asc_{PT}	0.83	−3.40	−5.12	−6.42	−4.07
asc_{Cycle}	1.71	−2.53	−3.02	−3.99	−3.63
asc_{Walk}	5.06	1.04	0.56	0.40	0.94
$\beta_{trav, Drive}$	−3.22	−5.90	−5.29	−5.96	−5.03
$\beta_{trav, PT}$	−6.96	−8.48	−5.72	−6.08	−7.80
$\beta_{trav, Cycle}$	−17.40	−10.88	−5.93	−6.23	−8.23
$\beta_{trav, Walk}$	−7.89	−6.57	−5.85	−6.12	−6.76
β_m	−0.05	−0.32	−0.60	−0.81	−0.30

more negative β_{trav} value for a specific mode indicates a stronger deterrent effect of travel time on the likelihood of choosing that mode. For instance, the impact of cycling travel time on transport mode selection is notably stronger for cohorts 1 and 2, whilst the influence of public transport travel time on the chosen mode of transport is stronger for cohorts 2 and 5. β_m represents the influence of monetary cost on the choice of travel mode which was also found to have a negative impact. As presented in Table 3, travel cost exerts a stronger influence on mode choice for cohorts 3 and 4 compared to other groups.

3.2. Model parameter and configurations

In this study we built our model based on the AToM workflow explained in Section 2 and further extended it. A 10% synthetic sample of the population of Greater Melbourne was generated using the activity-based transport demand model developed by Both et al. (2021). The network generation tool developed by Jafari et al. (2022) was used to create a road and public transport network for Greater Melbourne also described in Section 2. These were used as the main inputs of the simulation model.

The transport simulator component of AToM is based on the MATSim agent-based simulation toolkit (Horni et al., 2016). MATSim follows an iterative co-evolutionary optimisation process in which all agents

execute their daily travel plan using the transport network during each iteration. At the end of the iteration, agents score their executed plan using MATSim's utility-based scoring function. Agents with low scores are then selected with a high probability, and their travel plan gets slightly modified through arbitrary mutation. Route, travel mode or departure time are the dimensions that are usually considered for plan modification. This process of simulation, scoring and mutation is repeated for several iterations until the combined score of the population no longer improves and the system reaches equilibrium (Horni et al., 2016). The MATSim utility function parameters used in this paper are listed in Table 4 and are estimated based on the mode choice coefficients from Table 3 (see also Horni et al. (2016) for more details about MATSim's utility functions).

We defined driving and cycling as network modes, and utilised MATSim's queue-based simulator as the mobility simulator or function that allows agents to move around the simulation. Public transport was simulated using an implementation of the Round-Based Public Transit Optimised Router (RAPTOR) algorithm in MATSim, which in addition to being the most time efficient public transport router, also allows users to define the search radius used by agents within the simulation for finding public transport stops (Delling et al., 2015).

In this work, the initial search radius for finding a public transport stop was defined based on the average walking distance from home to a public transport stop among participants in the VISTA 2012–2018 travel survey. The analysis revealed an average walking distance of 1,600 m, indicating that individuals are generally willing to walk this distance to access public transport, similar to distances commonly used in the built environment field that most people would cover in an approximate 15–20 min walk (Jeffrey et al., 2019; McCormack et al., 2008). To account for variability in walking behaviour, the extended search radius was set to 1,900 m, reflecting the standard deviation of walking distances observed for home to public transport trips.

During the simulation model if agents were found to have access to at

least two public transport stops in the initial search radius, the search stopped, and the stop that minimised their travel time and cost would be selected. Otherwise, the search radius was increased incrementally by 1900 m each time until at least two public transport stops were found, from which one would be selected using the aforementioned criteria.

The MATSim sub-tour mode choice strategy was enabled for all four transport modes of driving, public transport, cycling, and walking; meaning that all agents could change their assigned transport mode between iterations to explore alternative modes that might improve their daily travel score.

We used a time allocation mutator within the MATSim module and altered the time mutation range, allowing agents to change their starting time by up to 2,700 s (or 45 min), to increase the likelihood that agents reach their destination on time. We modelled walking speed based on the average walking speed of 1.4 m/s as extracted from the VISTA travel survey data and using a distance correction factor of 30%. The distance correction factor increases the straight-line distance to approximate a distance that accounts for network restrictions.

3.3. Model calibration and validation

VISTA 2016–2018 data was used to calibrate the mode choice of agents for each of the four main travel modes. This comparison was done separately for each demographic cohort and for the total population. The calibration of the simulation model involved incremental adjustments to the alternative (mode-specific) constants for each travel mode across different demographic cohorts, as presented in Table 4. Our process used large incremental changes to the initial constants and as the error margin declined, smaller increments were used to ensure a comparable fit to the VISTA data. These constants account for unobserved factors influencing mode choice that are not explicitly included in the model. Calibration was conducted manually through an iterative process, with adjustments made within a predefined range to align the simulated mode share with the observed data. The calibration target was set to achieve a mode share within a $\pm 2\%$ error margin relative to the observed values. This approach ensured that the model was not over-fitted by avoiding exact replication of the observed mode shares, allowing for a reasonable level of variation while maintaining model validity. Fig. 2 shows the mode share comparison of the calibrated model with VISTA data.

The Root Mean Squared Error (RMSE) was calculated to evaluate the accuracy of the simulation model in replicating mode share distributions observed in the VISTA data across the five demographic cohorts. The average RMSE of 0.72 percentage points across the five demographic cohorts indicates that the simulation model demonstrates an overall good fit with the observed data, reflecting a relatively small average deviation between the simulation and actual mode shares. The model achieved the best fit for Cohort 4, with an RMSE of 0.48 percentage points, suggesting that the predicted mode shares closely align with the survey data for this group. Conversely, the highest RMSE of 0.99 for Cohort 3 indicates slightly more discrepancy between the simulated and observed mode shares. Overall, the low RMSE values across the cohorts confirm the model's reliability in representing real-world travel patterns.

To validate the use of public transport in the model, we compared the regional passenger flow share in our simulation model with Public Transport Victoria (PTV) Train Station Access data from 2016. The number of people using each of the 218 stations within the study region and the passenger flow share was aggregated to Statistical Area Level 3 (SA3)⁴ (ABS, 2016). We compared the passenger flow share of each SA3

Table 4
MATSim utility function parameters for each cohort.

Parameters	Mode	Cohort 1	Cohort 2	Cohort 3	Cohort 4	Cohort 5
Marginal utility of late arrival to an activity	–	–9.68	–17.70	–15.88	17.89	–15.08
Marginal utility of money	–	0.05	0.32	0.60	0.81	0.30
Marginal utility of performing the activity	–	3.22	5.90	5.30	5.96	5.03
Marginal utility of waiting for PT	–	–6.45	–11.80	–10.58	–11.93	–10.05
Alternative constant	Drive	0.00	0.00	0.00	0.00	0.00
	PT	0.83	–3.40	–5.12	–6.42	–4.07
	Cycle	1.71	–2.53	–3.02	–3.99	–3.63
	Walk	5.06	1.04	0.56	0.40	0.94
Marginal utility of travelling (per hour)	Drive	0.00	0.00	0.00	0.00	0.00
	PT	–3.73	–2.58	–0.43	–0.11	–2.77
	Cycle	–14.17	–4.98	–0.64	–0.27	–3.20
Monetary distance rate (AUD/1000 km)	Walk	–4.67	–0.67	–0.55	–0.16	–1.73
	Drive	–0.12	–0.12	–0.12	–0.12	–0.12
Daily monetary constant (AUD / day)	PT	–8.60	–8.60	–8.60	–8.60	–8.60

⁴ Statistical area level 3 (SA3) are geographical areas designed to provide a regional breakdown of Australia. They generally have a population of between 30,000 and 130,000 people. In major Australian cities, SA3s represent the area serviced by a major transport and commercial hub. SA3s are widely recognised as having a distinct identity and similar social and economic characteristics.

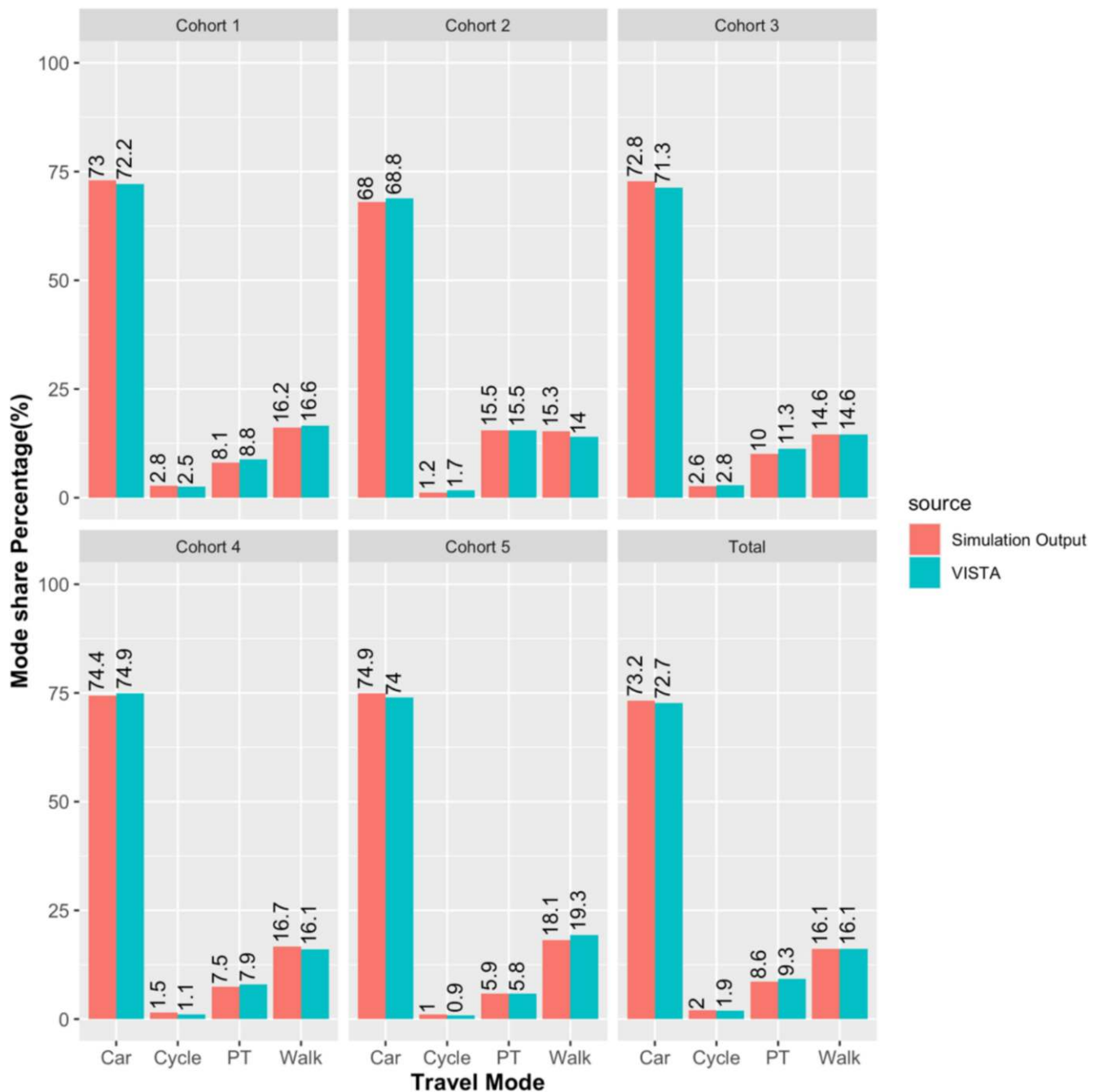


Fig. 2. Comparison of mode share percentage of the calibrated model to real-world VISTA data.

based on the aggregated PTV Train Station Access data with our simulation output. Fig. 3 shows this comparison. Rates of cycling in Melbourne are low, and due to a lack of data, we did not validate or consider cycling as a main mode of transport in this simulation model.

4. Multimodal train trip assignment

Public transport trips typically include other transport modes such as walking, cycling or driving, that facilitate access to or from public transport stops, described here as multimodal trips. As mentioned previously, the transport demand population of Both et al. (2021) focuses only on assigning trips' main mode and hence, MATSim will assign walking as the (main) travel mode for multimodal trips during the simulation. For home trips that use buses or trams, this is arguably inconsequential, as walking is the predominant mode of access for these

services. However, this is not necessarily the case for trips from home that involve train stations, since the percentage of people driving to train stations is considerable in the VISTA 2012–2018 dataset (at 54.5% overall), especially in the middle and outer suburbs of Greater Melbourne (ranging from 47% to 80%) where there are fewer transport options (Gunn et al., 2020). Trips to train stations by bike are considerably lower, with some authors suggesting that cyclists may prefer to cycle to their final destination rather than use cycling as a means of accessing train stations (Jeffrey et al., 2019), hence it is not considered in this section. Instead, it is crucial to capture driving and walking as separate and predominant access modes for trips to and from train stations to enable more realistic simulation modelling.

This section describes our process for creating the activity-based transport demand that explicitly includes trips between home and train stations by either driving or walking as transport modes for this

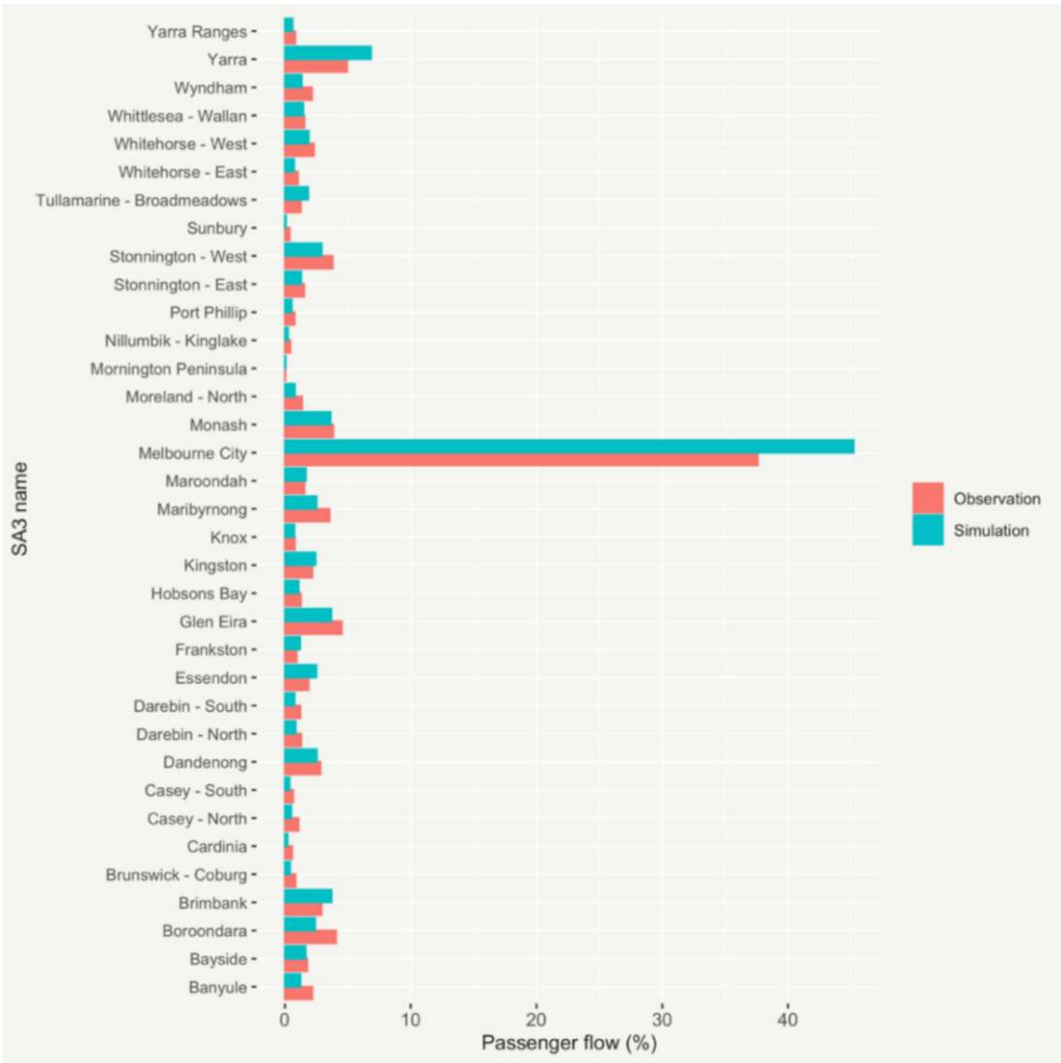


Fig. 3. Comparison of passenger flow for each SA3 in the simulation results and real-world observations.

subset of trips using the outputs of activity based transport demand model of Both et al. (2021) and the calibrated agent-based model. It should be noted that accessing train stations via other public transport modes (i.e., bus and tram) is captured during the simulation. Furthermore, the access mode from home to either a bus or tram stop is always modelled as walking for multimodal trips where the first public transport mode is either bus or tram.

4.1. Train station location assignment

We used the calibrated cohort-based multinomial models described in Section 3 to identify train users from the transport demand population and to identify train station locations that were used. For public transport trips, agents could choose between train, tram, or bus. The model was run for 250 iterations to ensure that all agents found their best transport mode and to allow those who chose public transport to find the best public transport stop to their destination in a way that would minimise the overall travel time and distance. Table 5 shows an example of the travel journey of an agent in the Both et al. (2021) transport demand population that used train as the first mode of transport. Table 6 shows the same travel journey after extending the demand model outputs of the activity-based transport demand model to add the missing steps for the intermediate travel between home and the train station.

The location of the train station, indicated as x and y coordinates, was assigned at this workflow stage. The transport modes for the access

Table 5
Travel journey of an agent in the original transport demand population.

Activity	Arriving mode	Act end time
Home	—*	09:50:00
Personal	PT — train	11:36:00
Shop	Walk	18:03:30
Home	PT — train	23:50:30

* The first activity of agents is always home where agents start their day, hence there is no arriving mode for this activity.

Table 6
Travel journey of an agent in the enhanced transport demand population.

Activity	Arriving mode	Act end time
Home	—*	09:50:00
Train Station	Car	09:52:00
Personal	PT — train	11:36:00
Shop	Walk	18:03:30
Train Station	PT — train	18:05:30
Home	Car	23:50:30

* The first activity of agents is always home where agents start their day, hence there is no arriving mode for this activity.

trips from Home to the Train station and from the Train station to Home,

along with the time that agents end their current activity and start travelling to their next activity indicated as *Act end time*, will be explained in the next section.

4.2. Home-train station travel mode assignment

To assign transport modes for trips between home and the train station (i.e., the Train Station Activity rows in Table 6), we first needed to identify candidates from the transport demand population that would potentially drive to train stations. To answer this question, we analysed the behaviour of train users within the VISTA 2012–2018 survey data. Trips from home to train stations were selected and classified into three regions of inner, middle, and outer Melbourne based on the home locations of the travellers. These regions were chosen as research suggests that these areas have distinct behaviours related to transport accessibility (Gunn et al., 2020; Jeffrey et al., 2019; Kroen et al., 2021). We then calculated the probability of driving from home to a train station for each region and the average driving distance to the train stations for each region. Table 7 shows the likelihood of driving to train stations and the average distances people are willing to drive by region. To improve the accuracy of assigning driving for trips to and from train stations, we considered region-based sampling, meaning we focused on two sampling measures specific to each region: the probability of driving to train stations and the distribution of driving distances. This approach aimed to enhance the precision of the spatial distribution among the selected agents.

We explored driving distance distributions by plotting a histogram using 500-metre bins. Various distributional shapes were examined to fit the probability density of the distance data. After evaluation, a log-normal distribution was determined to be the most accurate representation, showing the best fit for the observed data. This choice was substantiated by overlaying the log-normal distribution on the actual probability density, revealing a close fit (Fig. 4). The mean and standard deviation for each region's log-normal distribution are also included in Table 7. Table 7 shows that compared to Middle and Inner Melbourne residents, Outer Melbourne residents are more likely to drive as a transport mode and cover greater average driving distances for trips from home to the train station.

We took area-based samples of train users based on the measures described in Table 7 and deemed those to be candidates for driving, whilst the remaining train users were assigned walking as their mode for trips to and from train stations.

The activity end time was calculated based on Equation 3 in which $Act\ end\ time_{lag}$ is the end time of the previous activity, v is the speed (m/s) of the arriving mode, and d is the distance (m) travelled from the previous activity to the current activity.

$$Act\ end\ time = Act\ end\ time_{lag} + \left(\frac{d}{v} \right) \quad (3)$$

Table 7

Probability and distance distribution measures by inner, middle, and outer areas of Greater Melbourne.

Variable	Outer Melbourne	Middle Melbourne	Inner Melbourne
Probability of driving as a transport mode for trips from home to train station	0.80	0.47	0.12
Average driving distance to the train station (km)	3.67	1.94	0.94
Mean of distribution of driving distance to the train station (km)	1.41	0.98	0.63
Standard deviation of driving distance to the train station (km)	0.72	0.69	0.59

4.3. Simulation model summary

The final simulation model replaced the transport demand of the agent-based transport model described in Section 3, AToM, with the extended demand from Section 4, which now includes walking or driving trips between homes and train stations. Table 8 demonstrates the percentage of walking and driving trips before and after including multimodal trips (Section 4.2). The simulation settings remained the same as described in Section 3, except that the MATSim sub-tour mode choice strategy was not enabled for public transport to avoid mode change for the modified PT users.

5. The Suburban rail Loop stage 1 case study: SRL East

The Suburban Rail Loop (SRL) will provide a cross-city train line through the middle suburbs of Greater Melbourne. As originally announced, it contained three stages named SRL West, SRL North and SRL East, although SRL West has since been divided into SRL West and SRL Airport. In this section, we estimate travel pattern changes due to the introduction of the SRL East section (also called SRL Stage 1) in comparison to a Baseline scenario representing current mobility in Greater Melbourne.

The Baseline scenario uses the current road and rail network and population with demographic characteristics based on VISTA 2012–2018 and ABS Census 2016 data (ABS, 2016; State Government of Victoria, 2019). We did not consider new employment locations or other destinations that may become available in the future as the SRL is rolled out. Instead, by using the current demand of Greater Melbourne, we examined how the addition of SRL East will influence present mobility patterns, who is likely to be impacted, and where. The SRL East scenario was developed in accordance with the SRL business case (State Government of Victoria, 2021). We used the existing road and rail network from the Baseline scenario and added the six SRL East train stations to it. These six stations are located at Cheltenham, Clayton, Monash, Glen Waverley, Burwood, and Box Hill. Extra connecting public transport links were also added to connect these stations.

Fig. 5 shows the location of the SRL East stations and connected links. New hypothetical trains were allowed to operate between added SRL stations between 4.30am and 12.30am running every 6 min at peak times (7 am - 9 am and 4 pm - 6 pm) and every 10 min outside of these times. In measuring the impact of SRL East, we focused our analysis on the areas, or precincts, around SRL East train stations using 1600 m buffers in alignment with the SRL Business Case (State Government of Victoria, 2021). Hence, we analysed the behaviour of agents travelling to, from and between these areas as the focus of the case study since those living close to the new SRL East line are most likely to use it. We compared the simulation outputs between the Baseline and SRL East scenarios to capture changes in walking and driving as modes that affect health outcomes. Whilst cycling is also included in the modelling, results are not reported because cycling rates in Melbourne are low. More details on the modelling assumptions are described in the Supplementary material.

Results are presented in Tables 9 and 10. Table 9 focuses on trips between home and train stations and shows the percentage change in walking and car trips resulting from the SRL East scenario. It shows a modest increase in walking trips and a corresponding decrease in car trips resulting from mode shift away from driving and towards public transport. This shift occurs because the addition of new SRL East stations improve accessibility to the rail network, making public transport a more attractive option. As public transport trips inherently include access and egress components, travellers who previously drove directly now walk part of their trip.

Table 10 shows the percentage change in average daily walking distances of agents. It shows that the conversion of car trips to walking trips resulting from the SRL East scenario results in agents undertaking more walking (in place of driving) during their daily activities,

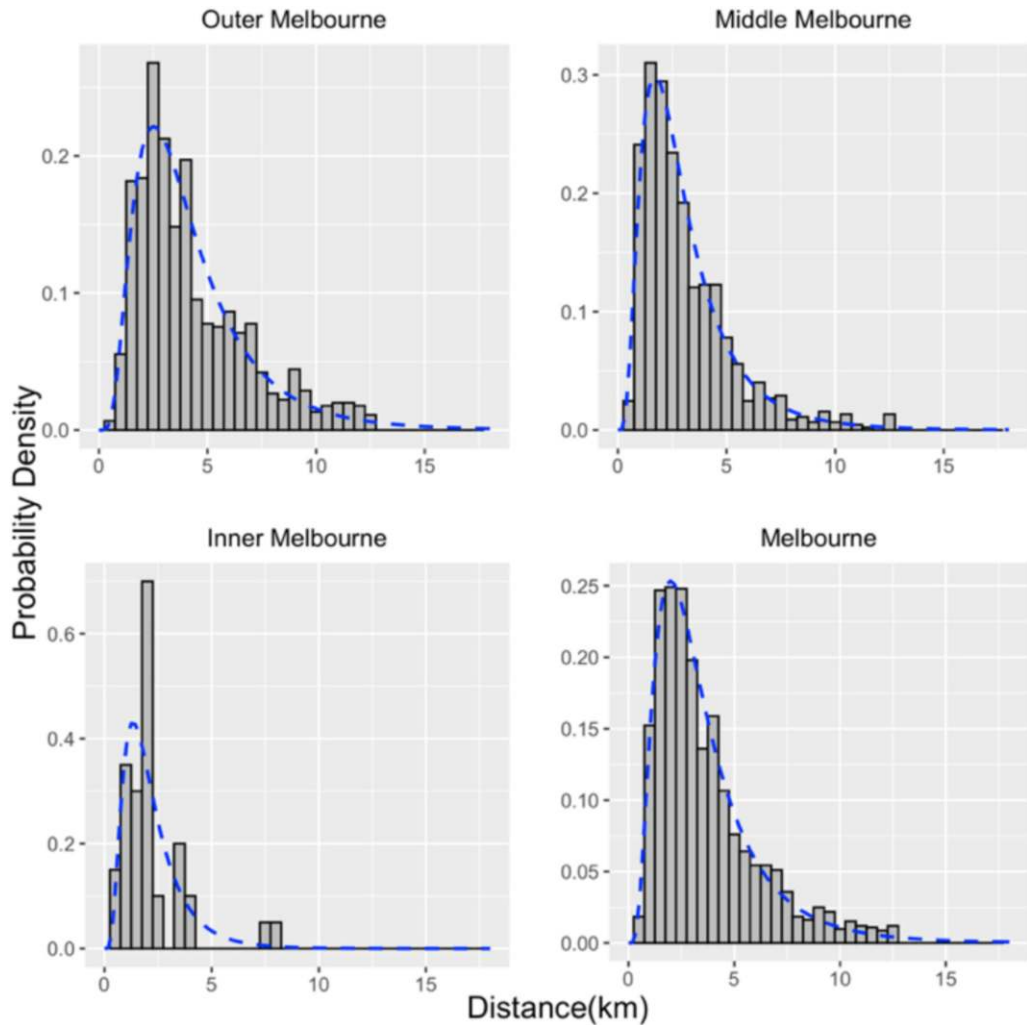


Fig. 4. Actual and fitted distribution of distance driven to the train stations for each region.

Table 8

Percentage of walking and driving to train stations in VISTA and in simulation outputs before and after explicit mode assignment for trips between home and station. Note: Percentages do not sum to 100% because other access modes are excluded.

Mode	Simulation output (Before) %	Simulation output (After) %	VISTA %
Drive	0	48	54
Walk	100	52	44

increasing the average daily walking distance by between 4.9% and 11.1% for the various cohort of agents. The greatest percentage change occurred for those in Cohort 2, who were males and females aged between 20 and 24 (11.1%). These increases are explained by the shift from car to public transport, which increases walking associated with public transport. Thus, the observed growth in walking distance reflects the behavioural impact of improved rail accessibility rather than a simple increase of walking distances.

The finding that the greatest percentile change in average daily walking distance occurs for Cohort 2 is explored further by the visualisation in Fig. 6. It shows the percentage increase in average distance walked for agents in each cohort (where there is an increase) as a result of the SRL East scenario, for agents residing in each Statistical Area Level 2 (SA2)⁵ (ABS, 2016) in eastern Melbourne in the vicinity of the SRL's location. Cohort 2 shows a particularly strong pattern of high increases in walking distances in the SA2s where the stations are located. It is the only cohort where five of the six SRL stations are located in or immediately adjacent to SA2s showing increases of over 10% in distances walked. Cohort 2 may be contrasted most strongly with Cohorts 4 and 5, where most of the SA2s where the SRL stations are located show either low or no increases in walking distances. For example, Cohort 2 is the only one where the distance walked in the SA2 containing Glen Waverly station (adjacent to a large commercial and retail hub) increases by more than 20%, and where the SA2 containing Burwood station (adjacent to a university campus) increases by more than 10%.

The cohort visualisations in Fig. 6 suggest a pattern where the SRL would be most effective in encouraging additional walking among

⁵ Statistical area level 2 (SA2) are medium-sized general purpose areas which, in urban areas, often represent a single suburb, though small suburbs may be clustered or large suburbs split into separate SA2s. They generally have a population of between 3,000 and 25,000 people, with an average of about 10,000 people.

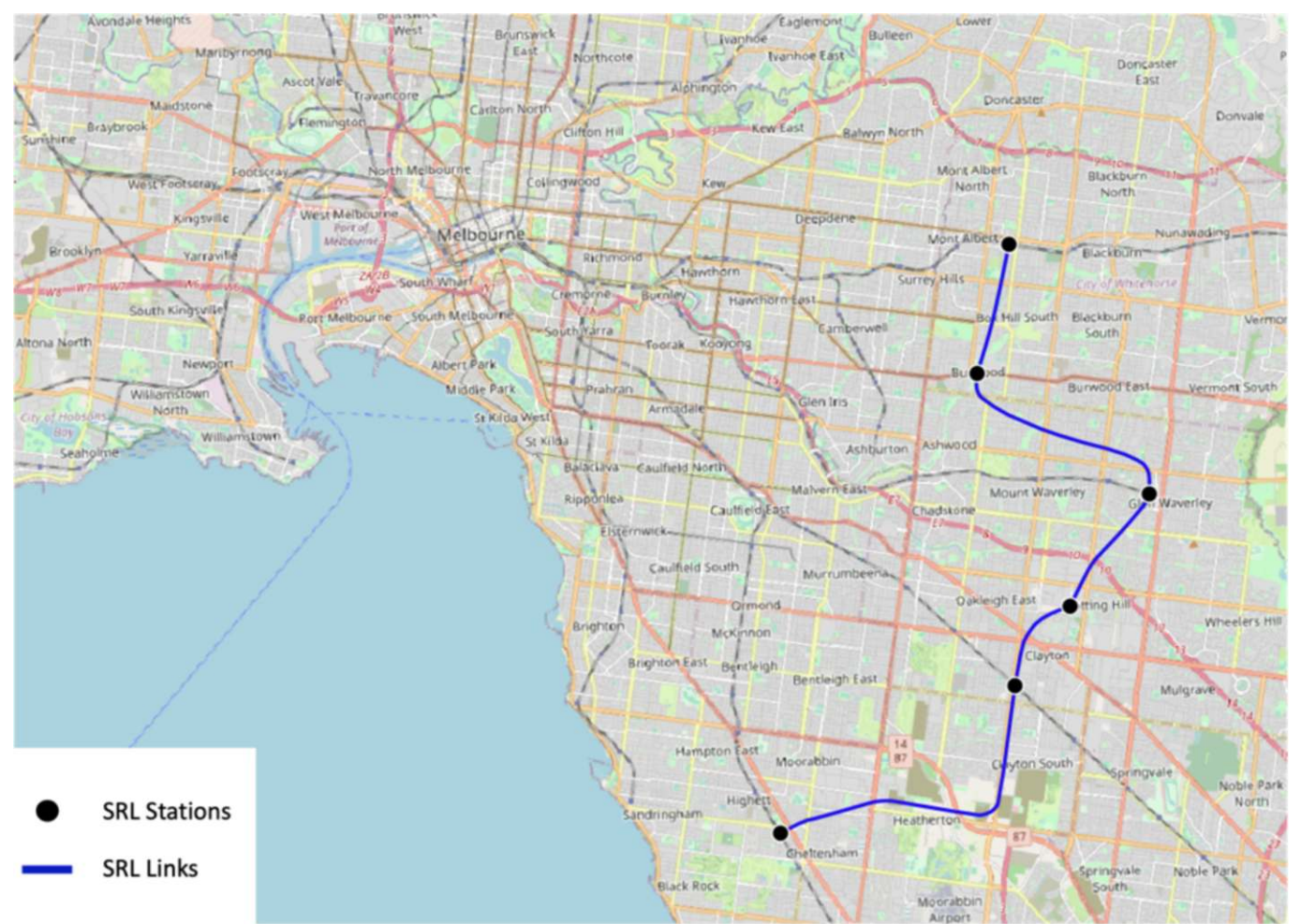


Fig. 5. Approximate location of the six stations included in SRL East (basemap © OpenStreetMap contributors).

Table 9
Percentage changes to walking and driving trips between home and train station, for trips to, from and between precincts.

Simulated trips	Baseline (%)	SRL East (%)	Change, percentage points
Walking trips	50.6	51.3	0.7
Car trips	49.4	48.7	−0.7

Table 10
Cohort-based percentage changes in average daily walking distance of agents travelling to, from or between SRL train station areas (%).

Name	Average daily walking distance increase (%)
Cohort 1	4.9
Cohort 2	11.1
Cohort 3	9.5
Cohort 4	5.1
Cohort 5	5.4

people in Cohort 2 living near the SRL stations, whereas for other cohorts the increases (which are lower) are more likely to occur for people living elsewhere but travelling to the SRL precincts.

The percentage of Cohort 2 agents in the SA2 containing Glen Waverley station (8.3%) is a little higher than the percentage of Cohort 2 agents in the simulation as a whole (7.6%), while the percentage of Cohort 2 agents in the SA2 containing Burwood station is much higher (17.2%), representing people of university student age residing near the

university. This breakdown indicates the kind of fine-grained information that can be extracted from the model, in this case indicating a station (Burwood) which is well situated in terms of nearby land use to improve walking activity among a cohort likely to undertake it.

6. Discussion

This research developed a workflow to enhance the existing AToM model. It includes several significant innovations including the development of mode choice models for different demographic cohorts of travellers based on age and gender. The cohorts were derived using clustering based on the approach of Both et al. (2021). Clustering is a commonly used approach in built environment literature (Both et al., 2021; Gunn et al., 2017; Jeffrey et al., 2019; McCormack et al., 2012), with the approach here developed using trip purposes of VISTA travel survey participants. The inclusion of age and gender via the cohorts used in the multinomial mode choice models allowed specific transport behaviours to be modelled enabling analysis of the effect of potential interventions on individuals and cohorts. We found females and males aged 20 to 24 (Cohort 2), along with females aged 25 to 29 and males aged 25 to 59 (Cohort 3) experienced the largest increase in average daily walking distance, increasing approximately 10% over the Baseline scenario. The results further indicate that there are more walking trips and a higher amount of walking for these cohorts. This is true even under the assumptions in this study that travel destinations will stay fixed and that transport behaviours follow existing patterns. A more significant shift towards active travel is likely beyond the modest changes found in

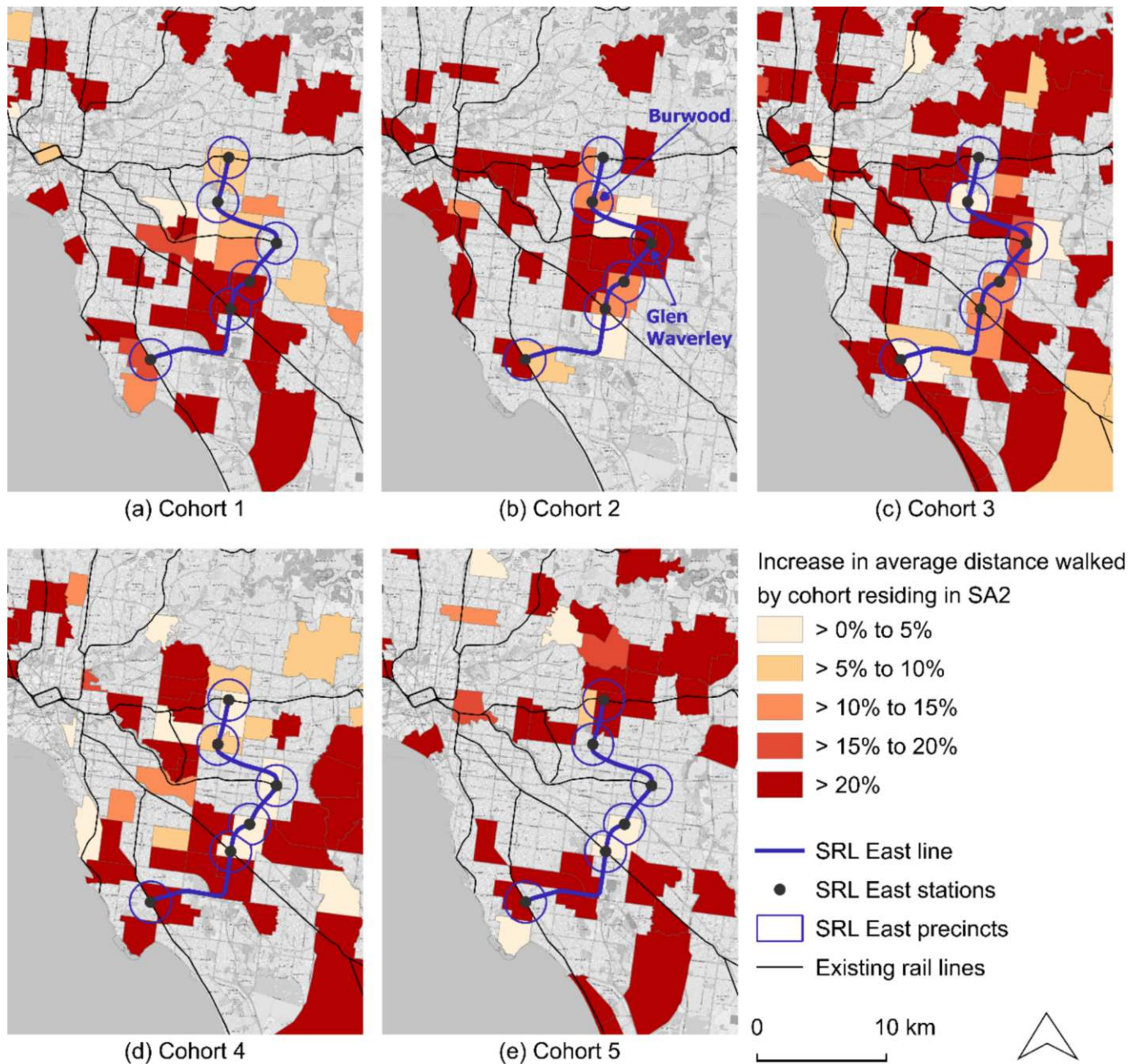


Fig. 6. Increase in average distance walked by cohort and Statistical Area level 2 (SA2) for agents travelling to, from or between SRL train station areas (basemap © OpenStreetMap contributors).

the SRL East scenario as the full SRL becomes operational, which includes the establishment of new lively precincts with jobs and other key local destinations such as shops and services, attracting more people to those areas to live and work. This information is useful for helping policymakers and planners create and improve scenarios tailored to the needs of specific population groups. For example, research evidence suggests that travel behaviours differ between men and women, and for households with children (Bautista-Hernández, 2020; Grue et al., 2020), yet the inclusion of age and gender has until now been excluded in mode choice models used within MATSim.

The multinomial mode choice models were developed to be parsimonious, and only included variables commonly used in transport literature such as travel time for all modes and the cost for driving and public transport (Frank et al., 2008; Ha et al., 2020). The use of these three variables penalised non-motorised modes such as walking and

cycling. Despite being cheaper, walking and cycling are slower and likely to be influenced by factors beyond time and cost such as individual preferences, which are not captured in the mode choice models due to insufficient data. In addressing this and to accommodate diverse and context-specific factors influencing mode choice, future research could draw upon more comprehensive travel behaviour and built environment data surveyed across different days of the week to establish common transport patterns linked to individual, social and built environment characteristics. For example, factors such as education, income, household structure, socioeconomic, and built environment factors including population density and destination accessibility can also influence mode choices (Boulange et al., 2017; Charreire et al., 2021). Furthermore, socioeconomic factors have been shown to impact transport mode choices both as individual and area-level factors and are worthy of exploration in future mode choice models (Ranjan & Sinha,

2025). These factors were not used in the present simulation as sufficient data was unavailable to adequately cover all combinations of individuals and their characteristics across modes and cohorts. This is also a function of the size of the underlying VISTA dataset and complexity of the multinomial mode choice model which requires sufficient data for robust and reliable estimation to support its predictive performance (de Jong et al., 2019). Their inclusion in future work will enhance the mode choice modelling and improve cohort development. Furthermore, alternate modelling approaches, such as expanded multinomial or nested logit models, could be used to explore a greater variety of transport mode options.

This leads to another innovation from the simulation modelling workflow. Our model can be expanded to include other transport modes, such as ferries, for trips from home to other public transport stops and *vice versa* making it applicable for use in other cities and contexts where urban sprawl, underserved, or car-oriented areas could benefit from more public transport. Furthermore, the modelling workflow is adaptable. For example, the demographic cohorts may differ across cities and contexts, and the mode choice modelling, contingent on data availability, could include other transport modes or variables beyond those modelled here. For example, GTFS and OSM data are readily available in other places including in Latin America (e.g. Mexico City, Buenos Aires), Africa (e.g. Nairobi, Addis Ababa), and South-East Asia (e.g. Ho Chi Minh City, Singapore) and the workflow can be adapted to cover missing data by initialising the models with similar or matched contexts following similar approaches taken by Kavta et al. (2024), who used mode choice models from Amsterdam and applied them to Manchester to examine electric mobility hubs, (Kavta et al., 2024), and Yasmin et al. (2015), who applied an agent based model from Toronto to Greater Montreal (Yasmin et al., 2015). In summary, use of agent-based models in data-poor contexts, could be based on the use of GTFS and OSM data plus the use of other relevant travel survey datasets or models from other cities or areas. A detailed exploration and framework development for this is left for future work.

Another innovation of this study is the extension of the model developed by Both et al. (2021) to simulate multimodal trips through walking and driving between home and train stations. This makes the model suitable for analysing train users' mode choice behaviour, understanding the behavioural impact of potential scenarios, and providing more accurate measures for distances travelled. For example, the SRL East scenario (Section 5) only investigated overall changes in walking distance of people who travelled within the SRL area. However, the model can capture the number of people who switched from driving to walking, and walking distance to train stations. Our analysis of the SRL East scenario found most trips between home and train stations in Greater Melbourne were undertaken by driving (over 50%) or walking (over 40%), whilst other modes such as cycling were less prevalent. However, this may not be the case in other cities or contexts, where cycling is more prominent.

Despite the various innovations of this simulation model, several limitations remain. We did not use any statistical tests to determine differences between the Baseline and SRL East scenario, nor did we undertake any sensitivity analyses owing to the simulation model run time being approximately five days. The study also relied on VISTA data from 2012 to 2018, prior to COVID-19, meaning more recent behavioural changes could not be reflected. Model calibration also required a manual process for mode constants. These limitations highlight directions for future work, including automated calibration methods, more comprehensive data collection, and sensitivity analyses to explore the impact on travel behaviour of different cost and mode choice parameters including the addition of a greater range of factors that influence mode choices (e.g., socioeconomic, built environment and individual level factors), travel behaviours and activity chains, infrastructure network changes and the testing of changes in the underlying population of people including their individual-level characteristics to accommodate future travel behaviours.

In our model, buses, trains, and trams effectively travel on separate networks based on already specified travel schedules. This means they are not affected by and do not contribute to road congestion. In Greater Melbourne, trams and buses predominantly share the road network with other motor vehicles, potentially impacting road traffic and travel times, but this is not captured in our simulation.

Furthermore, we only considered the inclusion of the SRL East section to the rail network, which is a small component of the full SRL that is ultimately intended to traverse many middle ring suburbs in Greater Melbourne. We did not attempt to model the full SRL as there was limited information on the exact route at the time of model development. The full SRL will likely generate even greater active and public transport impacts. Future research using agent- and activity-based simulation models could test the impact of the full SRL and other scenarios (e.g. increasing bus services or cycling infrastructure, land use changes) on travel behaviour.

We also did not investigate the role of demand management and specifically car parking as a mechanism for controlling transport behaviour, which may impact the decision to use a car as a mode of transport (De Gruyter et al., 2020; Manville, 2017). In some cities, restricting car parking at key locations has been a successful way to incentivise travel behaviour to modes other than car, however, the majority of this research focuses on car parking and demand management around universities, residential addresses and free street parking, but not train stations (Daggett & Gutkowski, 2003; De Gruyter et al., 2020; Ribeiro et al., 2020). Contrary to this, some authors suggest that car parking or park-and-ride facilities may support train use (Iseki et al., 2018). A detailed examination of car parking in the context of an agent- and activity-based simulation model is left for future research, since it would require consideration of car parking at both origins (e.g., residential locations) and at or near train stations as the interim destination. Furthermore, to increase walking in and around train stations or transit-oriented developments would require changes to land use and employment opportunities, which were not considered in the present research (Jeffrey et al., 2019). Specifically, changing the land use mix likely incentivises and increases SRL use, especially if jobs and key destinations are to be located along the new rail line.

This work focussed on model development and within this we defined several parameters that governed agent behaviour. We set the public transport fare for each trip at \$4.30 for all agents, which is realistic for this case study, however, public transport fares can vary, and concessions or discounts could impact agent behaviour through the utility scoring functions within MATSim. Furthermore, we treated the cost of driving as fixed and did not include all costs associated with cars such as the purchase and running costs of the car, insurance, maintenance, registration, or parking fees. As such train usage costs may be overestimated, whilst driving costs may be underestimated, especially considering recent increases in petrol prices and the cost of living. Future research could undertake sensitivity testing to establish the impact of these parameters in the modelling workflow. Finally, we did not consider any alternative or opportunity costs (or benefits) that could arise from the differing transport modes. For example, walking and cycling have physical and mental health benefits (Oja et al., 2011; Zapata-Diomedí et al., 2023), and train users benefit from social connection and can do other activities whilst travelling (e.g., reading, listening to music, or working).

7. Conclusion

Few transport simulation modelling studies use enhanced mode choice models that incorporate demographic information into a large-scale transport simulation model, and currently no studies calibrate across modelling components. This study used MATSim to extend an existing agent- and activity-based simulation model in addressing these research gaps. Our extended simulation model and workflow included a cohort-based multimodal mode choice model and accommodated trips

from home to public transport. Furthermore, we calibrated and validated the simulation model using VISTA data. Our extended simulation modelled walking, cycling, public transport, and driving trips using a case study scenario in Melbourne, Australia known as SRL East. Two scenarios were simulated: a Baseline scenario of the existing road and rail network; and the SRL East scenario, which supplemented the Baseline scenario by adding six new train stations constituting the proposed SRL East train line.

The model calibration results showed good concordance for each demographic cohort and their mode shares appeared to be well validated with minimal observed differences between train passenger flows derived from the simulation model when compared to PTV train station access data at the SA3 level. These results suggest the simulation model is sufficiently accurate to give realistic results when applied to real-world scenarios. We found that our SRL East scenario increased walking trips and average daily walking distances for the five cohorts of travellers, but especially amongst younger cohorts aged 20–24 who appeared to be amenable to increasing train use and walking within the SRL East case study scenario.

Future research could extend the mode choice models through the inclusion of a greater range of individual, social and environmental variables, additional modes, or through more complex regressions, such as multinomial or nested logits. The simulation model could also be tested through application to differing urban contexts.

To conclude, the simulation model and findings presented here are useful for developing infrastructure and behavioural interventions to support the uptake of public transport use and walking. Measuring these changes through a simulation model can help policymakers and planners understand the impact of proposed infrastructure in support of decision making and development for more sustainable and healthier cities.

8. Biographical note

The authors of this publication are members of the Healthy, Liveable Cities Lab in RMIT University's Centre for Urban Research. The Healthy, Liveable Cities Lab is a multidisciplinary team with expertise in public health, geographical information systems analysis, agent and activity-based modelling and statistical analyses of big data related to urban and transport planning applications.

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CRediT authorship contribution statement

Mahsa Abdollahyar: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Alan Both:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Afshin Jafari:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Steve Pemberton:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis. **Dhirendra Singh:** Writing – review & editing, Writing – original draft, Conceptualization. **Billie Giles-Corti:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization. **Lucy Gunn:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trip.2026.102018>

Data availability

Data will be made available on request.

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